

Nikhil J. Babani

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Education

Harvard University	Sep 2025 – Dec 2026
<i>Master of Science in Computational Science and Engineering</i>	Cambridge, MA
• Coursework: Manifold Learning, Numerical Linear Algebra, Stochastic Optimization, High Performance Computing	
Massachusetts Institute of Technology	Jan 2026 – May 2026
<i>Cross-registered Student</i>	Cambridge, MA
• Coursework: Algorithmic Trading, Portfolio & Risk Management, Options Market Making & Hedging, Statistical Arbitrage	
University College London	Sep 2020 – Jun 2025
<i>Master of Electronic & Electrical Engineering</i>	London, UK
• Grade: 1 st Class Honours (Top 5%)	
• Coursework: Data Structures & Algorithms, Object-Oriented Programming, Machine Learning, Deep Learning	

Experience

PwC	May 2026 – Aug 2026
<i>Incoming Software Engineer Intern</i>	Kuala Lumpur, MY
Khazanah Nasional Berhad	Jun 2025 – Aug 2025
<i>Software Engineer Intern</i>	Kuala Lumpur, MY
• Engineered an object-oriented Python backend for the External Fund Management team, using an inheritance-based class hierarchy to automate a legacy Excel scoring matrix used to evaluate 20+ equity portfolio managers' performance	
• Developed an internal, full-stack portfolio analytics dashboard for 10+ investment analysts within the Public Markets division to visualize manager performance and portfolio scorecards using Next.js, React, TypeScript, and FastAPI	
• Built a pytest validation suite using controlled inputs and reference Excel outputs to assert parity for the scoring matrix	
United Kingdom Atomic Energy Authority	Oct 2024 – May 2025
<i>Plasma Control Engineer Intern</i>	London, UK
• Simulated MAST-U vertical stability dynamics in MATLAB as part of a joint UCL–UKAEA control engineering project	
• Designed an MPC that improved transient and steady-state response, reducing average IAE and ITAE by 74.53%	
• Removed uncontrollable/unobservable eigenmodes and reduced computation time by 70.77% via minimal realization	

Projects

Parallel Monte Carlo Option Pricer C++, OpenMP, MPI, CUDA, JAX, Python, Slurm	Apr 2026 – Present
• Built a Monte Carlo European call option pricer across serial C++, OpenMP, MPI, CUDA, and JAX implementations, validating prices against the Black–Scholes solution and benchmarking performance on a Slurm-managed HPC cluster	
• Achieved 30.6× speedup with OpenMP on 36 CPU threads, 53.2× with MPI across 72 ranks on 2 nodes, 126.2× with CUDA on a single NVIDIA L4 GPU, and 52.8× with JAX GPU for 100M-path Monte Carlo workloads	
Image Classification for Medical Diagnosis Python, PyTorch, scikit-learn, Optuna	Dec 2024 – Jan 2025
• Built a PyTorch image-classification pipeline for BloodMNIST that handled tensor preprocessing, train/validation/test splits, and multi-class classification over 11,959 28x28 RGB training images across 8 blood-cell categories	
• Implemented and tuned hyperparameters of a compact CNN using Optuna, AdamW, early stopping, and learning-rate scheduling, reaching 94.56% test accuracy while improving classification robustness on underrepresented classes	

Skills

Languages: Python, MATLAB, R, C/C++, TypeScript/JavaScript, HTML/CSS, SystemVerilog, RISC-V Assembly, SQL

Frameworks & Libraries: React, Next.js, FastAPI, Pydantic, Tailwind CSS, Pandas, NumPy, scikit-learn, PyTorch, JAX

Tools & Platforms: Git/GitHub, CI/CD (GitHub Actions), Linux/UNIX, Bash, Jupyter, Slurm, CMake, VTune, Nsight Systems